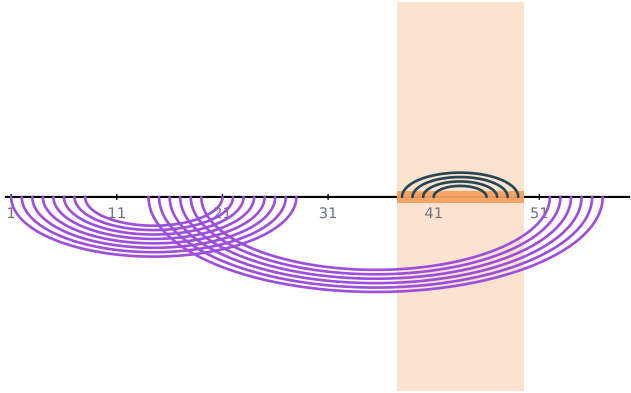
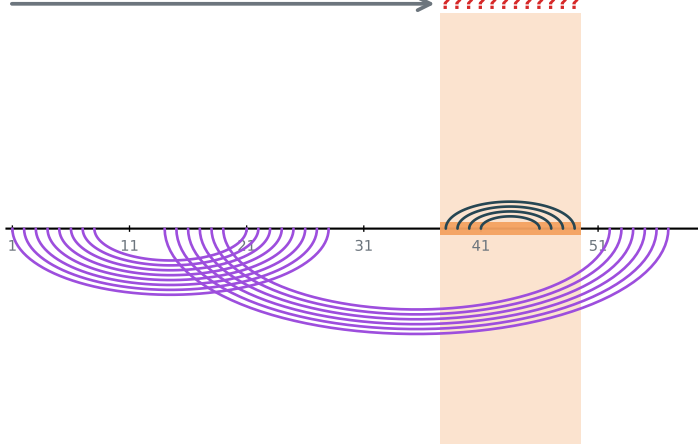


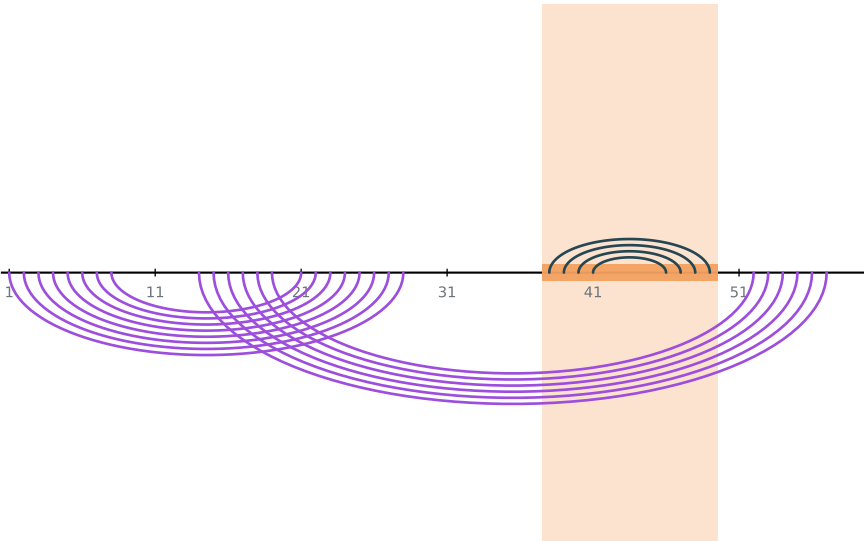
(a) Target: PreQ1-II switch (L=59 nt). Hold the orange hairpin (12 nt) to wild-type; design the rest.



(b) AR L→R: must commit to position 1 with no knowledge of the downstream fixed motif.
L→R decoder → **× incompatible** ???????????



(c) Order-agnostic (ours): random-permutation decoding places the motif first and designs around it. ✓ Jaccard = 1.00



design: GGAGGUGGAAAACGAAGGCACCACCUCCAAAAAAGCGGAUAACCGCAAGCCUUCAC

Fixed motif (WT) Nested base pair Pseudoknot base pair